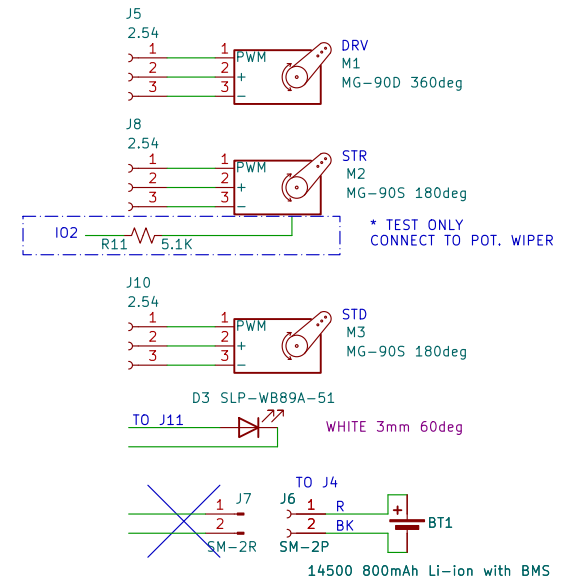
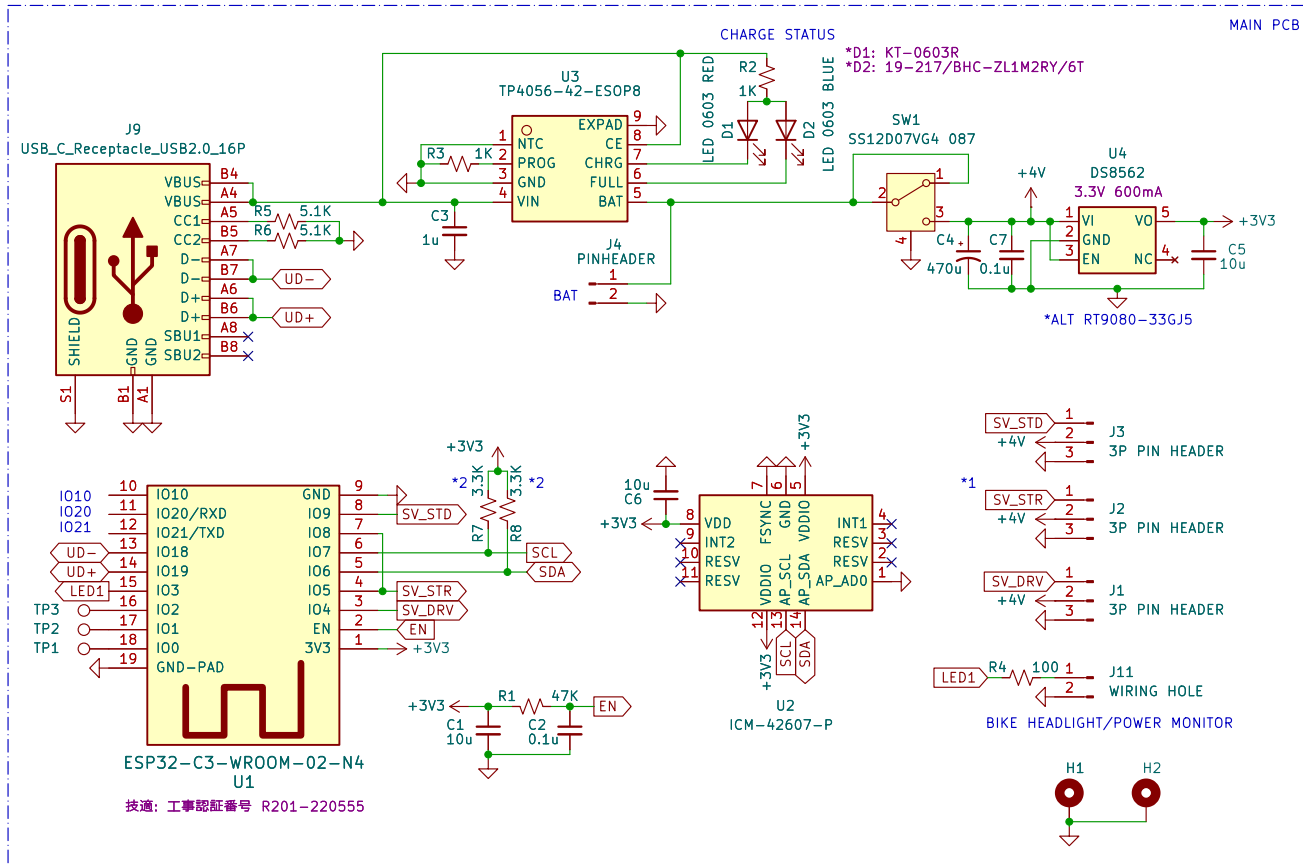
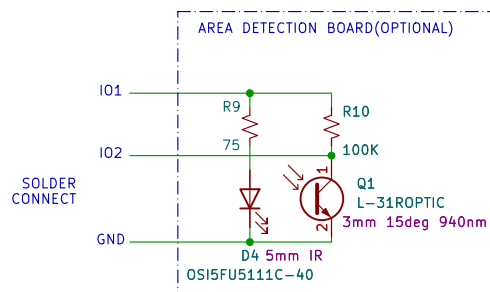




1. SLEEP CURRENT=16.8uA/6.0mA(NO SV./WITH SV.)
2. Battery 14500/800mAh
energy density = $3.7V \times 0.8Ah / (3.14 \times 1.4cm^2 / 4 \times 5cm / 1000mL/L)$
= 2.96Wh / 7.693mL = 385Wh/L < 400Wh/L PSE not needed.
3. Battery capacity measured = 500mAh at load = 5.1ohm
4. LED current = $(3.3 - 2.56) / 100 = 7.4mA$
5. MG-90S Minimum operation voltage = 3.1V(measured)
6. @3.8V Current = 95mA(WiFi connected)/ 600mA(str. locked)/ 215mA(rot. ON)/ 250mA(rot. + Oscillate str.)
7. @3.8V Deep sleep current = 6.9mA
8. Changed from 04: Add clearance between +4V and H2PAD.
9. CHG SW1
- 2025.12.20 CHG LED3
- 2026.01.11 CHG R7,R8 SHOULD BE 5.1K
- 2026.01.12 CHG R5,R6 REWORKED TO 5.1K
- 2026.01.16 ADD OPTION FOR AUTO PILOT
- 2026.04.30 ADD OPTION FOR TRAVEL ENCODER
- 2026.05.05 CHG R7,R8 REWORKED*2 TO 3.3K
- 2026.05.24 RMK MINIMUM OPERATING VOLTAGE=3.00V(MEASURED)



MAIN/SUB PCB : DOUBLE SIDED, FR-4, t=1.2mm

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Sheet: /
File: ROBOBIKE05_PCB.kicad_sch

Title: ROBOBIKE 05

Size: A4 Date: 2026-07-06

KiCad E.D.A. 10.0.1

Rev: 20251205

Id: 1/1